TAXI ROUTE OPTIMIZATION
WITH REINFORCEMENT LEARNING

A Q-learning agent learns pickup and drop-off decisions through rewards, penalties and experience.

PY Python · GYM Gymnasium · NP NumPy · MPL Matplotlib

Henil Patel - 202503103520166
Eric Patel - 202403103510248
Dev Lad - 202403103510287

1 TAXI WORLD: THE TASK

The taxi must navigate the grid to the destination.



6 actions: South ↓ North ↑ East → West ← Pick up Drop off

Conceptual illustration - not an exact screenshot of Taxi-v4.

2 HOW THE AGENT LEARNS

A feedback loop turns trial-and-error into a better policy.

1 OBSERVE State → 2 CHOOSE Action → 3 RECEIVE Reward → 4 UPDATE Q-table

THE Q-LEARNING UPDATE

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[r + \gamma \max Q(s',a') - Q(s,a)]$$

Explore: try actions • Exploit: choose the best known action

3 REWARD SYSTEM

Rewards tell the agent which outcomes are useful.

+20 **SUCCESSFUL DROP-OFF** — Task completed

-10 **INVALID PICKUP / DROP-OFF** — Avoid illegal actions

-1 **ORDINARY STEP** — Prefer shorter journeys

THE LEARNING SIGNAL

More reward over time usually means better decisions.
Always check delivery success and steps as well.

4 PROJECT SETUP & TRAINING

Configuration taken from your project guide.

20,000	500 × 6	$\alpha = 0.1$	$\gamma = 0.99$
training episodes	states × actions	learning rate	discount factor

Exploration ε 1.00 → 0.01

Exploration gradually decreases (decay = 0.9995).

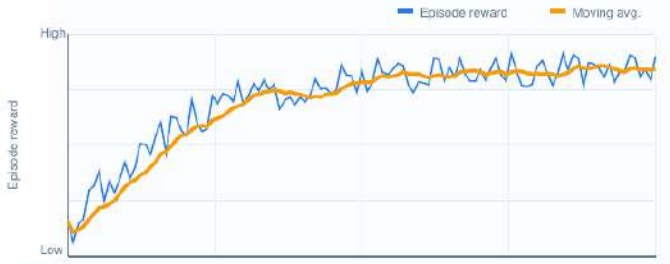
TRAINING PIPELINE

1 Reset new episode → 2 Act explore / exploit → 3 Learn update Q-value → 4 Repeat many episodes

PROGRAM OUTPUTS q_table.npy • learning_curve.png

5 VISUALIZING LEARNING

A learning curve shows how episode reward changes during training.



SCHEMATIC GRAPH - replace with your actual learning_curve.png

State	S	N	E	W	Pick	Drop
0	-0.2	0.1	0.4	-0.1	-1.2	-0.8
25	0.1	0.3	0.6	0.0	-1.0	-0.4
100	-0.1	0.2	0.5	0.4	-0.9	0.8
250	0.0	-0.2	0.4	0.2	-1.1	1.0
499	-0.3	0.1	0.2	0.6	-0.7	1.2

Values are examples for visual explanation, not the saved Q-table.

6 EVALUATION & NEXT STEPS

Use your real terminal output here - do not estimate.

AVERAGE REWARD _____

AVERAGE STEPS _____

SUCCESSFUL TRIPS _____ / 100



FUTURE SCOPE

Deep RL · Live traffic · Multi-taxi

SMARTER AGENTS • BETTER DECISIONS • MORE EFFICIENT ROUTES